

- 6 (a) Show that the curve with Cartesian equation

$$\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

has polar equation $r = \cos \theta$. [3]

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The curves C_1 and C_2 have polar equations

$$r = \cos \theta \quad \text{and} \quad r = \sin 2\theta$$

respectively, where $0 \leq \theta \leq \frac{1}{2}\pi$. The curves C_1 and C_2 intersect at the pole and at another point P .

- (b) Find the polar coordinates of P . [3]

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- (c) In a single diagram sketch C_1 and C_2 , clearly identifying each curve, and mark the point P . [3]

- (d) The region R is enclosed by C_1 and C_2 and includes the line OP .

Find, in exact form, the area of R .

[6]

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